

Overview

The goal of this project is to explore patterns of **urban sustainability and air quality** across major world cities. You will clean, merge, and analyze several open datasets to understand what factors explain changes in air quality and how clusters of cities differ in sustainability profiles.

You may work **individually or in pairs (1–2 students)**. Please indicate your choice by signing up for your ‘group’ on Canvas. Do this if you are working alone or in a pair so that you can submit the ‘group assignment’ properly. If working as a team, also include both names in your final submission.

Deadline: *March 6th 11:59pm*

Submission: A single PDF should be submitted to the Final Project assignment on Canvas. Your code as a .R, .py, .rmd, .qmd, or .ipynb file should be submitted to the Final Project Code Assignment on Canvas. Your pdf submission should stand alone. That is, I should not need access to your code to understand your report. Your report should also contain no code. Imagine that you are submitting this report to a future employer.

Part A – Data Wrangling

1. Data Collection

- Obtain data covering different aspects of sustainability including air quality, economic and social indicators, environmental productivity, population and urbanization:
 - OpenAQ Note: requires use of an API
 - World Bank Indicators
 - OECD Green Growth Indicators
 - UN World Urbanization Prospects
- You do not need to use all of these sources and you are not limited to only these sources. They are meant as a guide.
- You may gather data at the country or city level. You should try to gather data across a (at minimum) 10 year period.
- Be sure to gather a measure of air quality. I suggest PM2.5, but you may choose another if you wish.

2. Merging and Cleaning

- Standardize city or country names (you may need fuzzy matching).
- Merge data on country (or city) + year.
- Handle missing values by interpolating each variable within your time period using any methodology you consider appropriate (e.g., mean, median, mode, linear interpolation, spline, model-based).
- Clearly explain all choices in your write-up.

3. Urban Sustainability Index (USI)

- Construct a composite **Urban Sustainability Index (USI)** from multiple dimensions (environmental, social, and economic) based on your data. You get to choose how to define this USI as well as what dimensions to use and how much each dimension should affect the final USI
- You may use a weighted average, normalization scheme, principal-component-based index, or another method that you find fitting. Explain your reasoning.
- Include a table or figure summarizing the resulting variables for several example cities/countries/years.

Resources to Inform Your Urban Sustainability Index (USI)

You are encouraged to ground your index in established sustainability frameworks:

- **UN Sustainable Development Goal 11:** <https://sdgs.un.org/goals/goal11>
Defines measurable targets for housing, transport, air quality, and resilience.
- **OECD Green Growth Indicators:**
<https://www.oecd.org/greengrowth/green-growth-indicators/>
Groups sustainability metrics into four pillars: productivity, natural assets, quality of life, and economic opportunity.
- **World Bank Urban Sustainability Framework (2018):**
<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/745121519399645850/the-urban-sustainability-framework>
Describes six dimensions of sustainability: governance, fiscal health, environment, infrastructure, equity, and resilience.

Part B – Exploratory Data Analysis (EDA)

- Explore trends in air quality and sustainability indicators across time and regions.
- Include at least three high-quality visualizations
- Write a short narrative highlighting key insights and anomalies.

Part C – Predictive Modeling

- Predict **PM2.5** (or another air-quality measure) using multiple predictors.
- Compare at least three regression models.
- Discuss:
 - Data preprocessing steps (scaling, normalization, encoding).
 - Model architectures or hyperparameters.
 - Evaluation metrics
- Summarize your findings in a concise paragraph and visualization

Note: We talked mostly about classification models. You must generalize what you've learned to regression.

Part D – Dimensionality Reduction for Regression

- Reduce predictor dimensionality and compare regression performance using reduced vs. full feature sets.
- Provide a 2-D and/or 3-D visualization of the reduced representation and explain the variance explained or latent structure learned.

Part E – Dimensionality Reduction for Unsupervised Clustering

- Use only the predictor columns (not the air-quality outcome).
- Cluster cities in a reduced-dimensional space (you may want to try a few dimensionality reduction techniques)

- Visualize clusters in 2-D/3-D and discuss whether they correspond to meaningful regional, economic, or environmental patterns.

Part F – Cluster Analysis of Outcomes

- Compare average **PM2.5** and **USI** across your clusters.
- Interpret differences among groups (“clean but unsustainable,” “dirty but improving,” etc.).
- Summarize in a short paragraph with one clear visualization.

Part G – Reflection on Modeling and Overfitting

- Suppose you chose a metric of interest and trained hundreds of models to maximize your chosen metric. Discuss the potential dangers of this approach and how to mitigate them.

Evaluation Rubric

You will be evaluated based on:

- Use of best practices as discussed in class.
- Quality of reporting, including explanations, documentation, reproducibility, and readability.
- Quality of the analyses and the results.
- How challenging the analyses are in terms of innovative thinking and exploration of methods and ideas.
- You should aim to use mostly models that we have learned in this class. Though you may use LASSO or Ridge Regression in Part C. You should also use at least 2 neural networks, whether they be regression models, autoencoders, or both.

Category	Description	Weight
Data Wrangling & Documentation	Merging, interpolation, construction of USI, clarity of methods.	20 %
Modeling & Evaluation	Appropriate choice and comparison of models, explanation of metrics.	40 %
Visualization & Interpretation	Quality and readability of figures; depth of interpretation.	30 %
Clarity & Reproducibility	Organization, clear writing, reproducible code, well-structured submission.	10 %